

The Impact of AI-Supported Digital Health on Efficiency in Mental Health Services: A Scoping Review

Quijada-Fernandez, David; **Kang, Heimin**; Jung, Jiwon; Kleinsmann, Meike; van Os, Hine

Introduction: The global shortage of mental health professionals and rising demand for care have intensified interest in digital health interventions (DHIs) and artificial intelligence (AI) to improve efficiency in care provision. While AI-supported DHIs show promise in enhancing access, engagement, and clinical decision-making, evidence on their economic efficiency and cost-effectiveness remains limited. Given the growing popularity of AI, there is a need to assess how AI-supported DHIs influence (in)efficiency in mental healthcare services. Therefore, this scoping review aims to map the impact of AI-supported DHIs on (in)efficiency in mental health care provision and highlight research gaps and opportunities for digitalisation in mental health services.

Methods and analysis: The review follows the scoping review methodology by Arksey and O'Malley, expanded by Levac et al. and the Joanna Briggs Institute. A search strategy using the concepts of mental health, AI-supported DHIs, and efficiency was designed to consult the following databases: Embase, Medline, Cochrane, PsycInfo, Web of Science, and Google Scholar. Two reviewers will independently screen and select studies against inclusion criteria, and one reviewer will conduct data extraction and synthesis, with a second reviewer extracting data from a random sample of the included studies to ensure rigour.

Keywords: *Artificial intelligence, Efficiency, Digital health, Mental health, Psychiatry*

Expected Contributions

- Investigates how AI-supported digital health interventions influence both efficiency and inefficiency in mental health service delivery.
- Identifies conceptual and methodological gaps in how efficiency is defined, operationalized, and measured in existing research on AI in mental health services.
- Provides directions for the future design, implementation, and evaluation of AI-supported interventions to strengthen sustainable and efficient mental health service delivery.